BY

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INTENDED LEARNING OUTCOMES (ILOS)

By the end of this lecture students should be able to:

- Explain the epidemiology and determinants of rabies
- Recognize modes of transmission, incubation period and clinical features
- Understand diagnosis and case management
- Evaluate prevention and control strategies

INTRODUCTION

Rabies is an acute, almost 100% fatal viral infection of the central nervous system.

It is caused by Rabies virus (Lyssavirus).

It is primarily a zoonotic disease transmitted from animals to humans.

Death usually occurs due to respiratory failure.

GLOBAL EPIDEMIOLOGY

Occurs in more than 150 countries worldwide.

Most deaths occur in Asia and Africa.

Children under 15 years account for about 40% of exposed persons.

More than 15 million people receive post-exposure vaccination annually.

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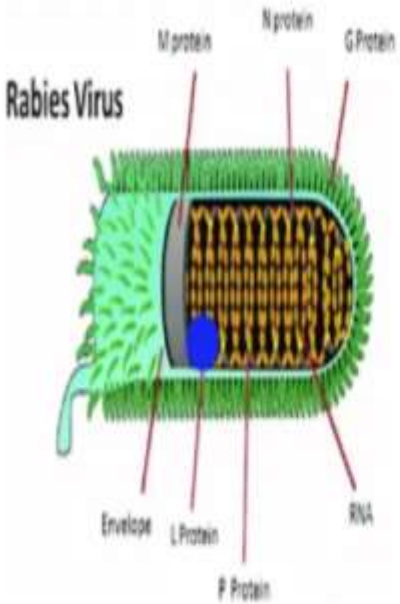
ETIOLOGICAL AGENT

Rabies virus

Family: Rhabdoviridae

Genus: Lyssavirus

A neurotropic virus affecting the central nervous system.



Rabies

RESERVOIR AND SOURCE OF INFECTION

Reservoir: Many wild and domestic animals.

- Dogs (responsible for up to 99% of human cases)
- Foxes, wolves, coyotes, jackals
- Skunks, raccoons, mongooses

Source of infection to humans: saliva of rabid animals.

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HOST FACTORS

All warm-blooded animals including humans are susceptible.

Rabies in humans is a dead-end infection.

High-risk groups include:

- Veterinarians
- Laboratory workers
- Dog handlers
- Hunters and field naturalists.

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MODE OF TRANSMISSION

Main route: Bite of a rabid animal.

Virus is transmitted through saliva.

Entry may occur through:

- Bite wounds
- Scratches
- Broken skin or mucous membranes

Rare: organ transplantation or aerosol exposure.

INCUBATION PERIOD

Usually 8 –3weeks.

Influenced by:

- Site of bite (distance from brain)
- Severity of wound
- Amount of virus inoculated
- Host immunity.

CLINICAL MANIFESTATIONS

- ***Early symptoms:***
- • Fever
- • Headache
- • Malaise
- • Pain or tingling at bite site

- ***Progressive symptoms:***
- • Hydrophobia
- • Aerophobia
- • Delirium and convulsions
- • Paralysis leading to death.

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DIAGNOSIS

Clinical diagnosis based on history of animal bite and symptoms.

Laboratory confirmation may include:

- Antigen detection (immunofluorescence)
- Virus isolation from saliva
- Detection of viral RNA.

TREATMENT AND CASE MANAGEMENT

No specific curative treatment once symptoms appear.

- Isolation in quiet room
- Sedation and pain relief
- Maintenance of hydration
- Respiratory and cardiac support.

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PREVENTION AND CONTROL – ANIMAL MEASURES

Control of rabies in animals is essential.

Measures include:

- Registration and vaccination of dogs
- Control of stray dog populations
- Active surveillance of animal rabies cases.

PREVENTION – PRE-EXPOSURE PROPHYLAXIS

Recommended for high-risk groups.

Vaccination schedule:

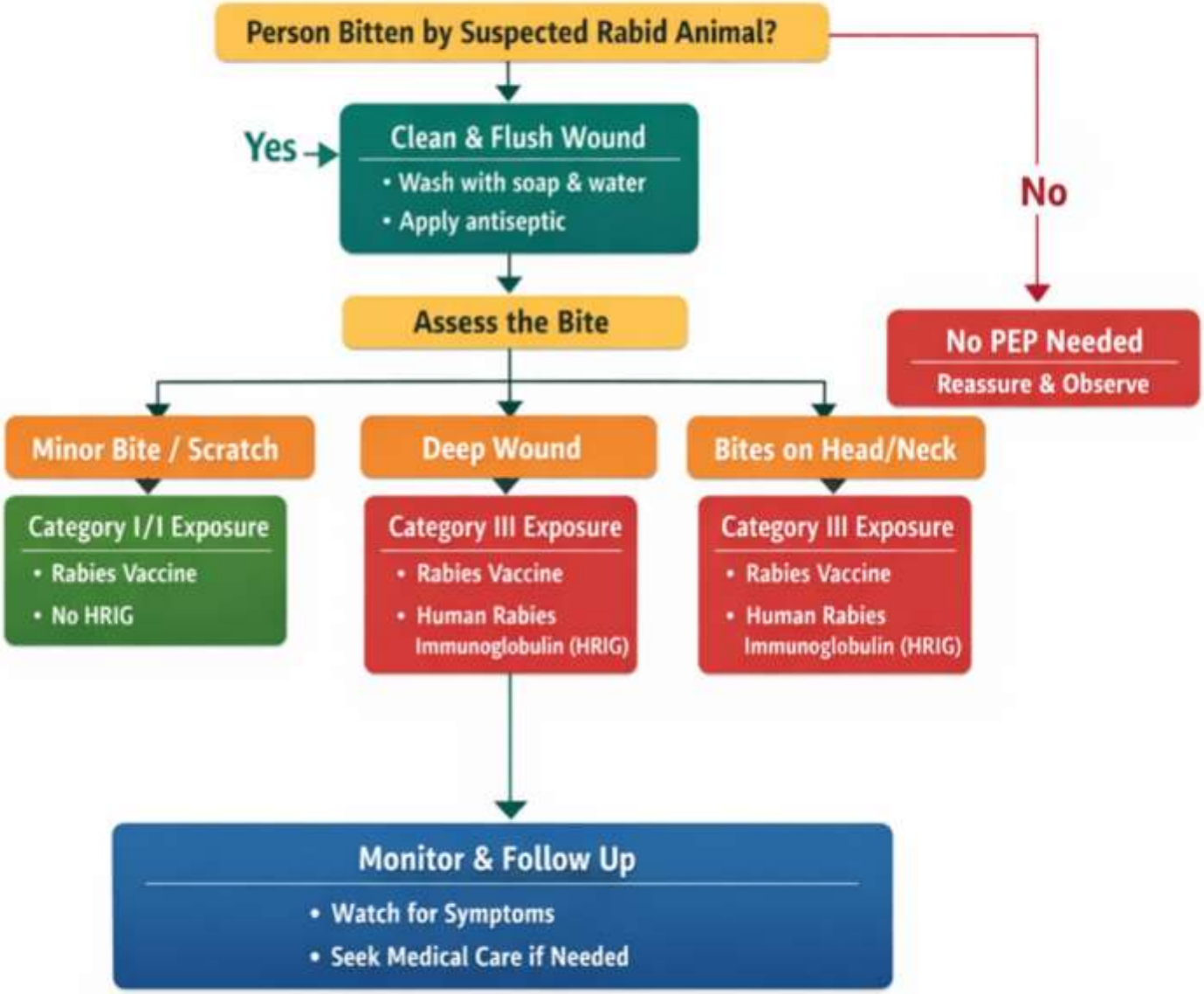
3intramuscular doses at days 0,7,21,28.

Booster doses recommended if exposure risk continues.

POST-EXPOSURE PROPHYLAXIS

- Immediate wound washing with soap and water.
- Rabies immune globulin (HRIG).
- Rabies vaccination.
- Tetanus prophylaxis and antibiotics if necessary.

Rabies Management



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🐶 15-YEAR-OLD MALE BITTEN BY HIS DOMESTIC DOG AND TRANSFERRED TO HOSPITAL.

• 1 MEASURES TOWARD THE PATIENT

Immediate Wound Management

1. Wash the wound immediately with soap and running water for at least 15 minutes.
2. Apply antiseptic (iodine or alcohol).
3. Do not suture the wound unless absolutely necessary.

2. Rabies Post-Exposure Prophylaxis (PEP)

a) Rabies Immunoglobulin (HRIG)

Dose: 20 IU/kg body weight.

Infiltrate around and into the wound as much as possible.

Remaining dose given intramuscularly at another site.

b) Rabies Vaccine

Given intramuscularly according to schedule: Day 0,3,7,14,28

3. Other Supportive Measures

Tetanus prophylaxis if not previously immunized.

Antibiotics if the wound is contaminated or deep.

Proper medical observation and follow-up.

- 2 **Measures Toward the Dog**

1. Observation

The dog should be confined and observed for 10 days.

2. Veterinary Assessment

Examination by veterinary authorities.

3. If the Dog Shows Signs of Rabies

The dog should be euthanized.

Brain tissue examined in laboratory for rabies virus.

4. If the Dog Remains Healthy for 10 Days

The dog is considered not infectious at the time of the bite.

Public Health Advice

Vaccinate domestic dogs regularly.

Report suspected rabid animals to veterinary authorities.

Avoid contact with stray or aggressive animals.

KEY PUBLIC HEALTH MESSAGE

Rabies is preventable.

Prompt post-exposure prophylaxis can prevent disease.

Mass vaccination of dogs is the most effective control strategy.

IT'S QUIZ TIME

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Post-Test MCQ 1

- A child is bitten by a suspected rabid dog. What is the most important immediate measure?
- A. Start antibiotics
- B. Immediate wound washing with soap and water
- C. Wait for symptoms
- D. Isolation of the patient

Post-Test MCQ 2

- What is the most common source of rabies transmission to humans worldwide?
- A. Cats
- B. Bats
- C. Dogs
- D. Rodents

Post-Test MCQ 3

- What is the usual incubation period of rabies?
- A. 1–2 days
- B. 3–8 weeks
- C. 6 months
- D. 1 year

Post-Test MCQ 4

- Which measure is most effective for rabies elimination?
- A. Isolation of patients
- B. Vaccination of dogs
- C. Antibiotic prophylaxis
- D. Water sanitation